

Name _____ Date _____ Period _____

Writing the Inverse of a Linear Function

Algebra II | Unit 1 | Day 11 | TEK A2.2.B

Objective: I can write the inverse of a linear function algebraically by swapping x and y and solving for y , then check my answer by graphing or with a table.

Warm-Up — You Already Know How to Do This

In Algebra I you solved literal equations for a variable. That is the whole skill today.

Solve $y = 3x + 6$ for x : $x =$ _____

The Three Steps

Step 1. Replace $f(x)$ with _____.

Step 2. _____.

Step 3. Solve for y , then write it as _____.

Part 1 — Worked Examples

Example 1. $f(x) = 2x + 3$

Swap: _____ Solve: _____

So $f^{-1}(x) =$ _____

Example 2. $f(x) = 3x - 5$

Swap: _____ Solve: _____

So $f^{-1}(x) =$ _____

Part 2 — You Try

1. $f(x) = x + 7$

$f^{-1}(x) =$ _____

2. $f(x) = 4x$

$f^{-1}(x) =$ _____

3. $f(x) = 5x - 10$

$f^{-1}(x) =$ _____

4. $f(x) = -2x + 8$

$f^{-1}(x) =$ _____

Part 3 — Check It Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$ from Example 1.

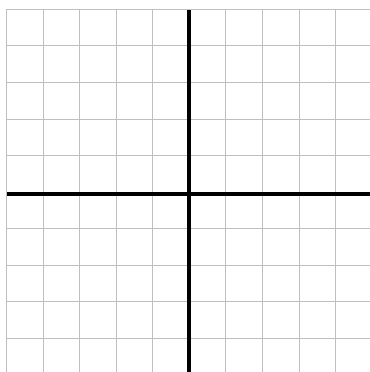
Check 1: the tables should trade columns.

x	-1	0	1	2
f(x)	1	3	5	7

x	1	3	5	7
f⁻¹(x)				

Check 2: plot both. They must mirror across $y = x$.

Graph f and f^{-1} , plus the line $y = x$.



Record it.

Slope of f : _____

Slope of f^{-1} : _____

What is the relationship?

y-intercept of f is 3. Of f^{-1} ?

Part 4 — Watch Out

A student says the inverse of $f(x) = x + 4$ is $f^{-1}(x) = 1/(x + 4)$. What went wrong?

Which linear function is its own inverse? _____

What is the inverse of a horizontal line, like $f(x) = 3$? _____

Part 5 — In Context

To convert Celsius to Fahrenheit: $F(c) = (9/5)c + 32$.

Find the inverse: $F^{-1}(x) =$ _____

What does the inverse do in words? _____

Practice

Write each inverse. Show the swap-and-solve steps.

1. $f(x) = (1/2)x - 6$

$f^{-1}(x) =$ _____

2. $f(x) = -x + 9$

$f^{-1}(x) =$ _____

Exit Ticket

Write the inverse of $f(x) = 6x - 12$, then verify it with one input.

$f^{-1}(x) =$ _____

Check: $f(3) =$ _____ and $f^{-1}(\text{_____}) =$ _____